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Final Technical Report

Kriti '26



AEROMODELLING MARUT DRONES

Hostel ID:

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1 Introduction

Drone based reforestation addresses one of the biggest challenges in environmental restoration: how to plant on a scale without sacrificing survival rates. Manual planting is precise but slow, while traditional aerial broadcasting where seeds are scattered in bulk is fast but wasteful. Many seeds end up clustered together or fall on unsuitable surfaces such as rocks, reducing their chances of survival. A drone system that releases seedballs one at a time, at controlled intervals, offers a more balanced approach combining efficiency with precise placement for large-scale planting efforts.

By dropping seedballs at regular intervals, each plant gets a equal share of sunlight, water, and nutrients from the soil, reducing unnecessary competition. Without this controlled spacing, seeds often fall in clusters, leading to overcrowded patches where only the strongest seedlings survive. At the same time, other areas are left empty, making them more prone to invasive species taking root.

1.1 Importance of timing and control

Consistent timing and accurate drop control are not just operational details they play an important role in ensuring meaningful and sustainable results. In drone based reforestation, the overall success of plantation efforts depends greatly on how precisely seed placement is managed during flight. When release intervals are uniform, plant spacing becomes more predictable, survival rates improve, and land coverage becomes more efficient.

1.1.1 Uniform Timing: It keeps the drone's speed and seed release perfectly in sync, so the drops form a pattern, regardless of wind or terrain.

1.1.2 Seed Drop Control: Guarantees that only one seedball is released at a time, preventing mechanical jams and ensuring that the seedballs are used efficiently.

1.2 Problem Definition

The primary challenge is to address the limitations of traditional manual reforestation—such as high labor costs and limited access to rough or protected terrains—by utilizing drone technology. Current aerial seeding solutions are often hindered by mechanical jamming caused by non-uniform seedball sizes and a lack of precision control over dispensing rates. Consequently, there is a need for a reliable and intelligent payload system that can autonomously manage seedball dispersal in remote environments.

1.2.1 Project Scope and Requirements

We are tasked with designing and developing a standalone seedball dropping mechanism to be mounted on a drone. The project focuses exclusively on the payload system; the drone's flight control and propulsion are out of scope.

1.2.2 Core Requirements:

1.2.2.1 Storage & Capacity: The system must be of up to 20 kg including both the mechanism aswell the seedballs within a maximum volume of 300 mm x 350 mm x 350 mm.

1.2.2.2 Material Handling: It must reliably process non-uniform, irregular spherical seedballs with diameters ranging from 15 mm to 35 mm.

1.2.2.3 Precision Dispensing: The mechanism must allow for adjustable drop intervals between 1 and 5 seconds.

1.2.2.4 Autonomous Intelligence: The system must include onboard electronics like ESP32 and sensors to detect and automatically rectify blockages or jams without human intervention.

1.3 Constraints

1.3.1 Mass Limits: The total allowable mass, including the dispensing mechanism and seedball payload, is capped at 20 kg.

1.3.2 Physical Constraints

Maximum payload volume: Width: 300 mm, Height: 350 mm, Length: 350 mm

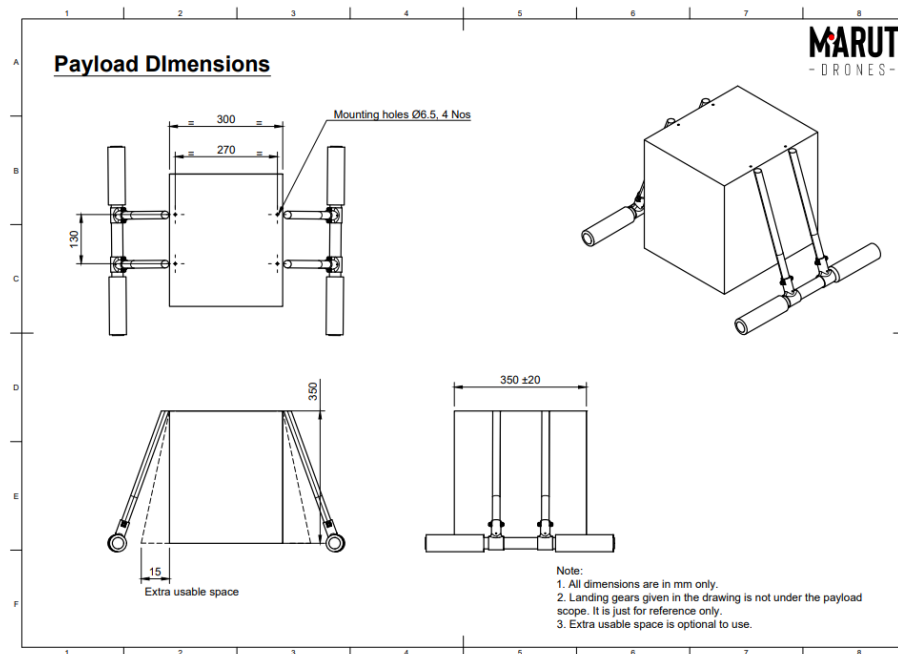
Seedball size range: Diameter: 15 mm to 35 mm

Shape: Non-uniform / irregular spherical

Weight optimization for aerial deployment

1.3.3 Controlled seedball drop interval: 1-5 seconds.

1.3.4 Vessel and Screw Placement



2 Conceptual Ideation

This section covers the evolution of design ideas explored for the seedball dispensing system, highlighting the reasoning behind different concepts and the criteria used to evaluate them.

2.1 Objective and Methodology

2.1.1 Thinking Structure : Working Backwards

To get a suitable design, we first examined what typically goes wrong in seed dispensing systems rather than assuming an optimal mechanism from the outset while also ensuring that the drop needs to be the same every

time, regardless of how full the hopper is.

- (a) **How do we stop seeds from clumping?** Seeds naturally want to stick together; we need to break that bond.
- (b) **How do we handle different sizes?** A mechanism for a 15 mm seed might crush or jam a 35 mm seed.
- (c) **How do we ensure only one drop?** We need a way to “queue” the seeds so they do not all rush out at once.

2.1.2 Concept Research Plan

The research was centered on existing industrial systems designed to solve similar problems.

- (a) **Seed Storage Vessels:** The research examined factors that contribute to arch formation such as particle size variation, wall friction, and hopper geometry and explored design strategies to minimize these issues.
- (b) **Vending Machines:** Examined spiral dispensing mechanisms commonly used in vending machines to understand how they isolate and release single items per rotation. The study focused on how rotational pitch, pocket spacing, and controlled motor actuation prevent multiple releases while maintaining consistent and jam-free operation.

2.1.3 Purpose of Mechanisms

We simplified the design by dividing the mechanism into basic components, each responsible for one clearly defined function.

- (a) **Auger:** A screw that moves a steady volume of seeds forward.
- (b) **Disc:** A rotating plate that catches one seed and drops it at a set interval.
- (c) **Agitator:** A stirrer that keeps the seeds moving so they don't get stuck in the hopper.

2.2 Design Constraints and Tolerances

The final mechanism was developed within clearly defined mechanical, operational, and environmental constraints to ensure reliable performance under real-world drone deployment conditions.

2.2.1 Failsafe Mechanisms

- (a) **Agitator:** Keeps the “pile” loose so the main screw always has seeds to grab, and prevents arching of seeds.
- (b) **Directional Reversal:** When a stall is identified, the controller initiates a brief reverse rotation of screw for 2% of seed dropping time period to clear the jam and restore normal seed flow.

2.2.2 Seed Tolerance Limits

We set a target range for seeds between 15mm and 35mm.

- (a) To minimize the risk of jamming, all internal flow paths are designed with some extra space usually around 1.15 times the diameter of the largest seedball. This allows for natural size variation, surface irregularities, and small shifts during motion. As a result, the seedballs move more smoothly through the system, lowering the risk of clogging or mechanical interference..
- (b) If the clearance is too small, friction and obstruction can increase to the point where the motor stalls or the mechanism jams. Conversely, if the gap is too large, more than one seedball can slip through at the same time, which affects the accuracy of single drops and disrupts the spacing consistency.

3 Mechanism

3.1 Engineering Logic & Design Philosophy

The seedball dispersal payload proposed aims to solve the major issues in aerial seeding like flow inconsistency, jamming, and seedball size variation while at the same time, focusing on reliable and autonomous operation within drone's weight, volume, and power limitations for remote, sensitive environments. It uses gravity assisted feeding combined with controlled mechanical transport to separate bulk storage from dispensing, thus, delivering consistent output across different fill levels and reducing blockages/ uncontrolled release to the minimum.

Using simple, modular, and commercial parts helps to increase manufacturability, maintenance, and scalability, thus, they strike a balance between engineering rigor and practical usability to ensure reliable performance.

3.2 Major Hardware Components

The payload consists of three primary mechanical subsystems. The hopper assembly serves as the seedball storage unit and incorporates a tapered geometry with integrated rotational agitation to ensure consistent feeding of non-uniform seedballs. The horizontal screw conveyor (auger) provides controlled, metered transport of seedballs from the hopper to the dispensing stage while acting as a buffer against flow surges. Final release is achieved through disc dispensing mechanism, where a continuously rotating dispensing disk, featuring conical radial grooves with varying incident angles, accommodates different seedball sizes for controlled and sequential discharge through a single release point.

3.3 Technical Specifications

The technical specifications of the proposed seedball dispersal payload are defined to ensure reliable operation within the mechanical, electrical, and spatial constraints of a drone-mounted system. Key specifications are summarized below.

(a) **Overall mass:**

- Maximum total payload mass (mechanism + seedballs): 20 kg.
- Designed with conservative margins to account for dynamic loads and vibration during flight.

(b) **Dimensional constraints:**

- Maximum length: 350 mm
- Maximum width: 300 mm
- Maximum height: 350 mm

(c) **Seedball Parameters:**

- Average seedball diameter: 32 mm (varying from 15mm to 35mm)
- Average seedball mass: 27 g

(d) **Hopper Assembly:**

- Gravity-fed hopper with tapered geometry
- Internal rotational agitator to prevent bridging and stagnation driven using DC geared motor

(e) **Screw Conveyor:**

- Horizontally mounted threaded screw for metered transport
- Driven by a stepper motor selected based on torque requirements

(f) **Dispensing Mechanism:**

- Disc with three conical grooves of different angular widths to accommodate seedballs of all varied sizes.
- The open groove collects the balls dropped by the screw and carries it till the dropping hole through its rotation

(g) **Control System:**

- Microcontroller-based architecture with ESP32

- Controls motor actuation, sensor inputs, and fault response logic
- (h) **Sensors and Monitoring:**
 - TMC2209 Motor stall sensing for screw and disc jamming detection
- (i) **Power and Electrical Interface:**
 - Powered directly from the drone's onboard battery
 - Includes voltage regulation, protection circuitry, and internal power distribution
- (j) **Structural Materials:**
 - Carbon Fiber : Lightweight with High Tensile strength
 - Acrylic Sheet : Easy for fabrication and gives optical clarity

3.3.1 Intended Payload

The proposed seedball dispersal payload is designed to operate within the constraints of a drone-mounted system while ensuring reliable mechanical and control performance. The total payload mass, including the mechanism and seedballs, is limited to 20 kg, and the system is packaged within approximate dimensions of 350 mm × 300 mm × 350 mm. It is designed to handle seedballs with diameters ranging from 15mm to 35mm, while accommodating reasonable geometric variation. The hopper features a gravity-fed tapered geometry with integrated rotational agitation to prevent stagnation. Seedball transport is achieved using a horizontal screw conveyor, and controlled dispensing is provided by a disc mechanism with varied grooves for multiple seedball dimensions. A microcontroller-based system manages actuation, sensing, and fault detection, with power drawn directly from the drone's onboard supply.

3.3.2 Detailed CAD Views of Components

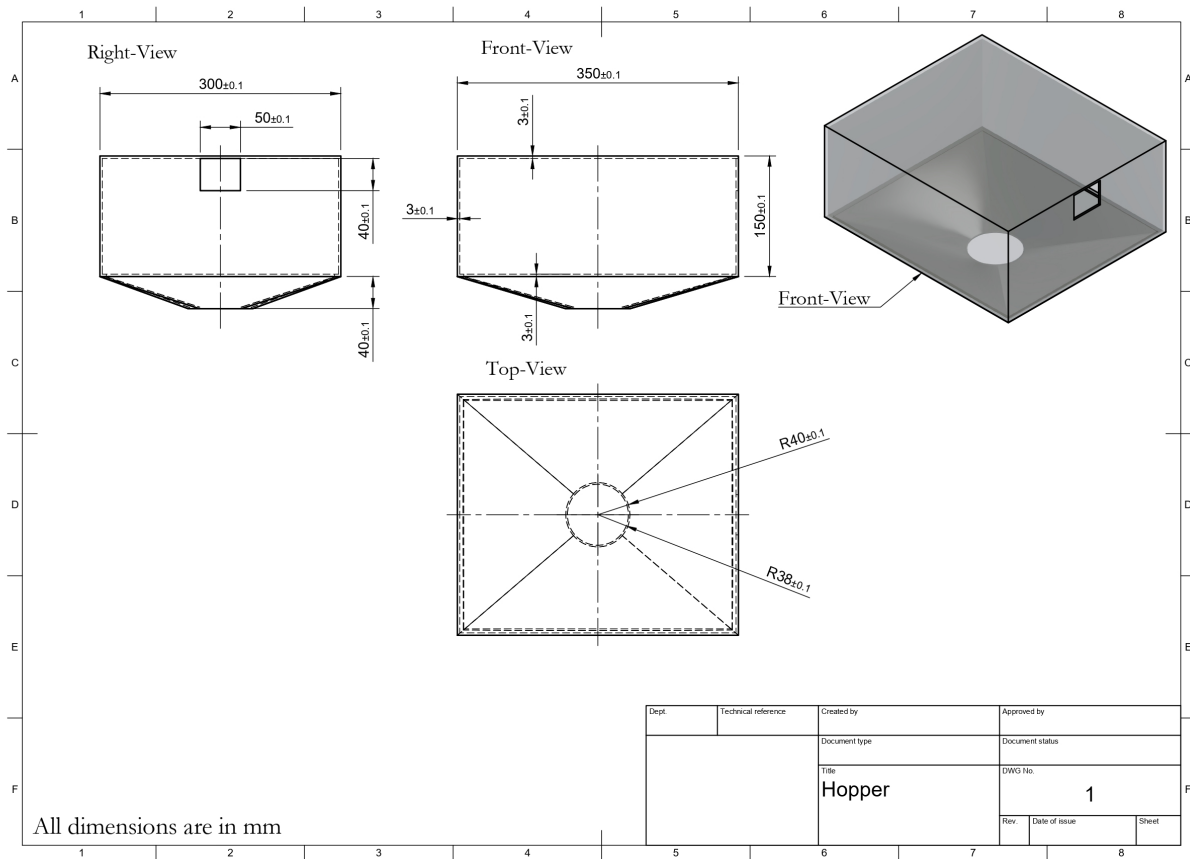


Figure 1: Hopper

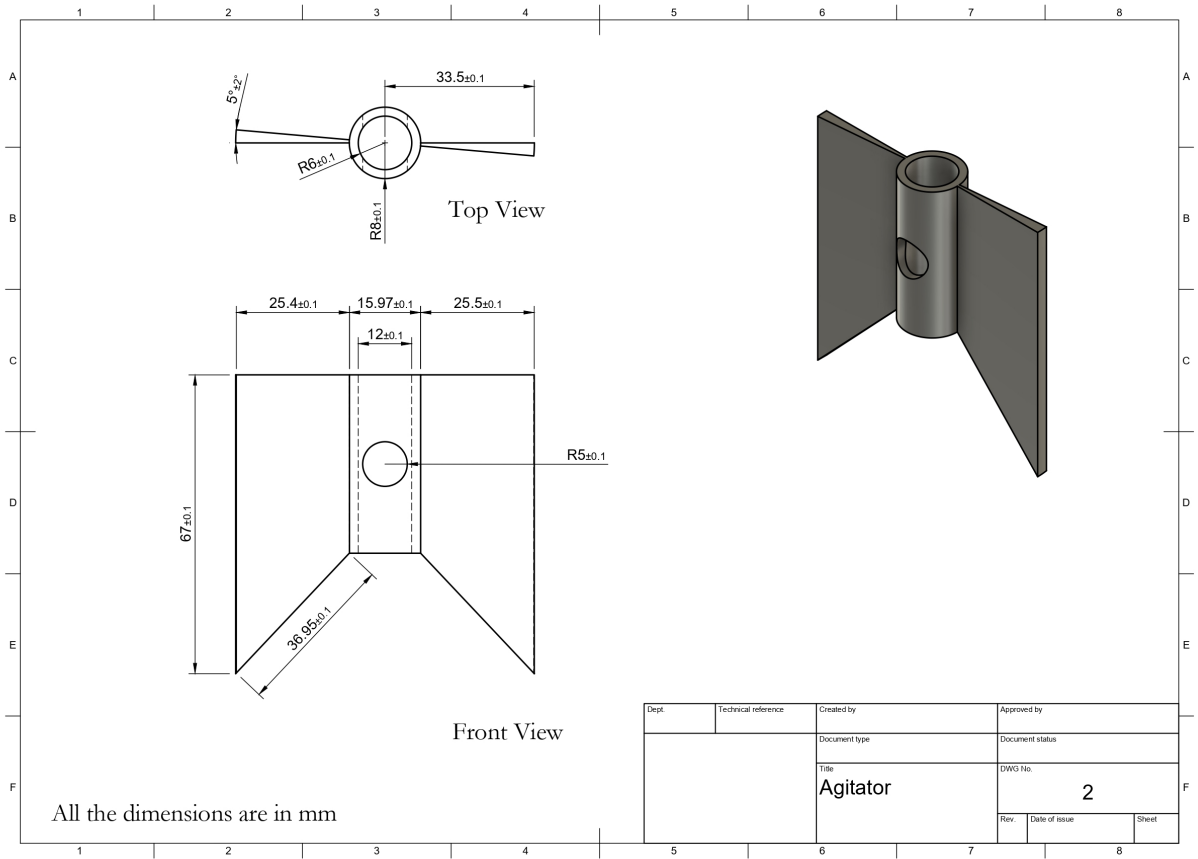


Figure 2: Agitator

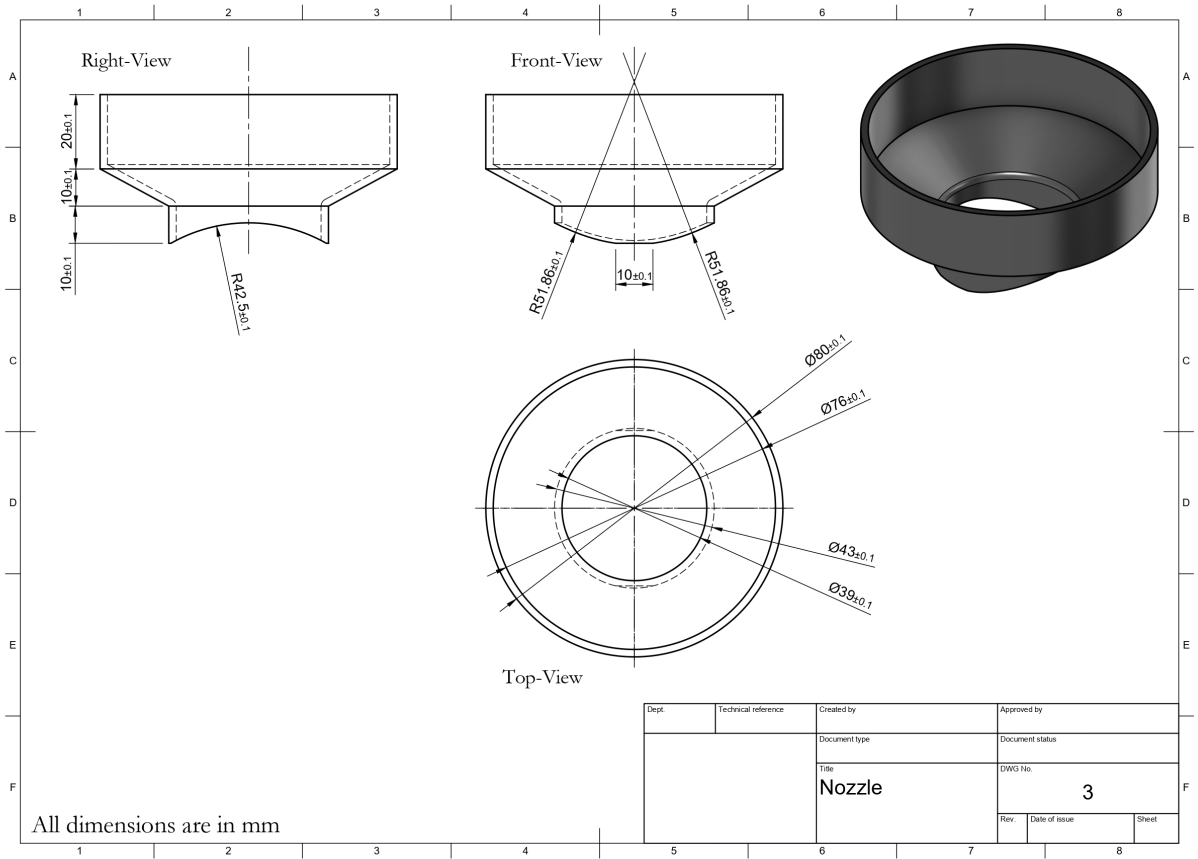


Figure 3: Hopper Nozzle

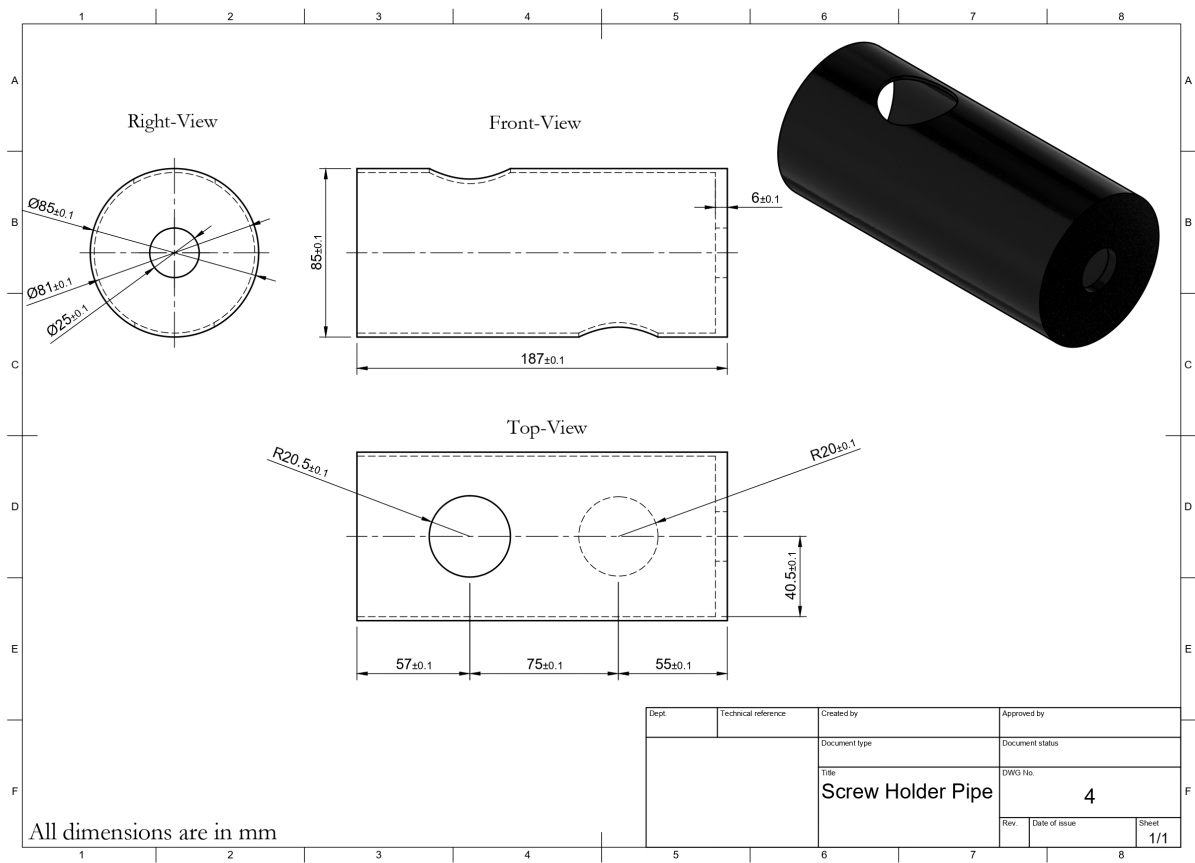


Figure 4: Screw Holder

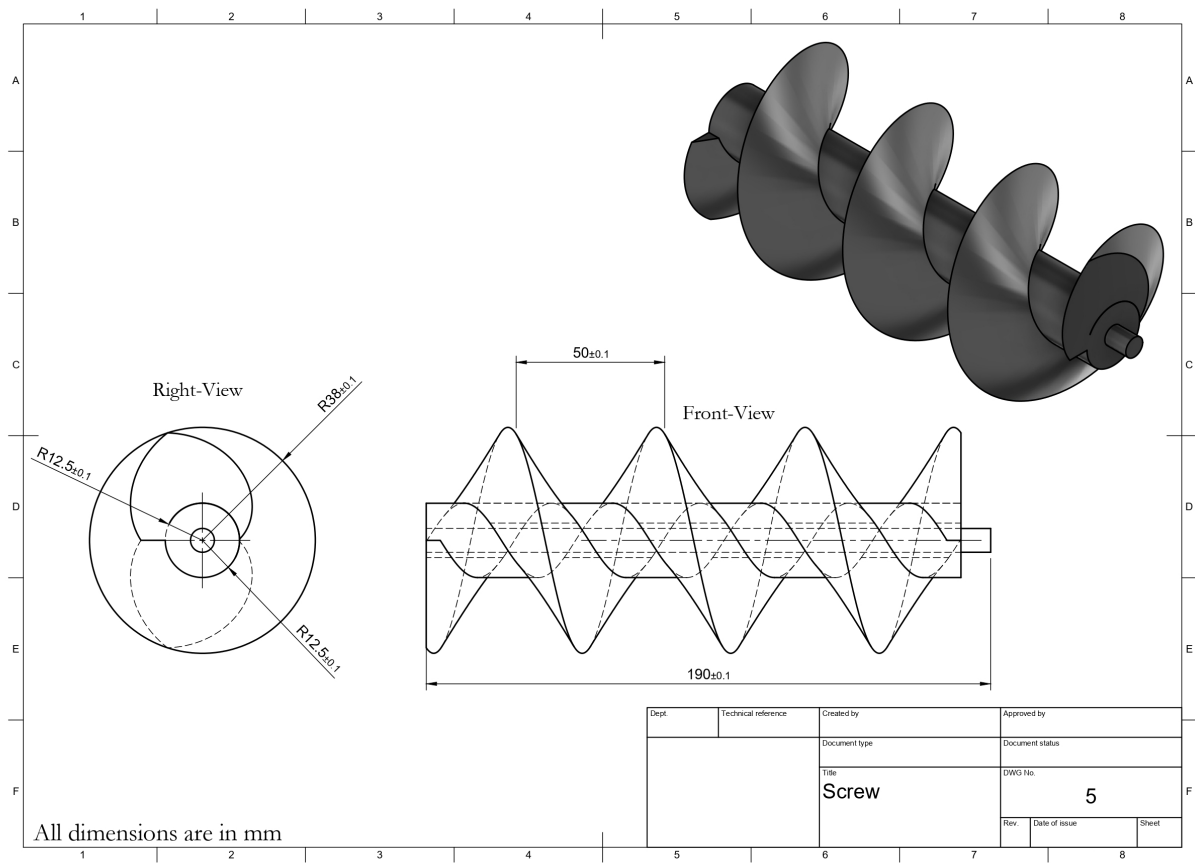


Figure 5: Screw

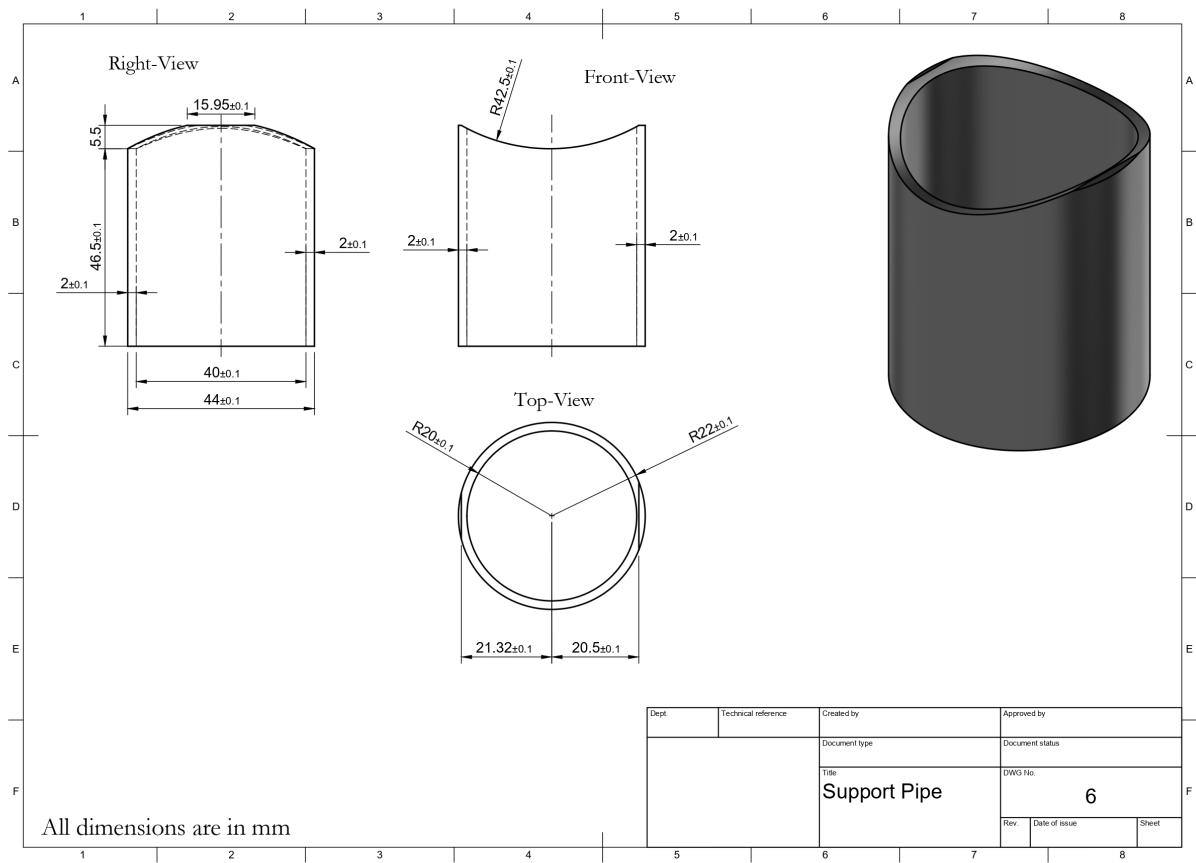


Figure 6: Support Pipe

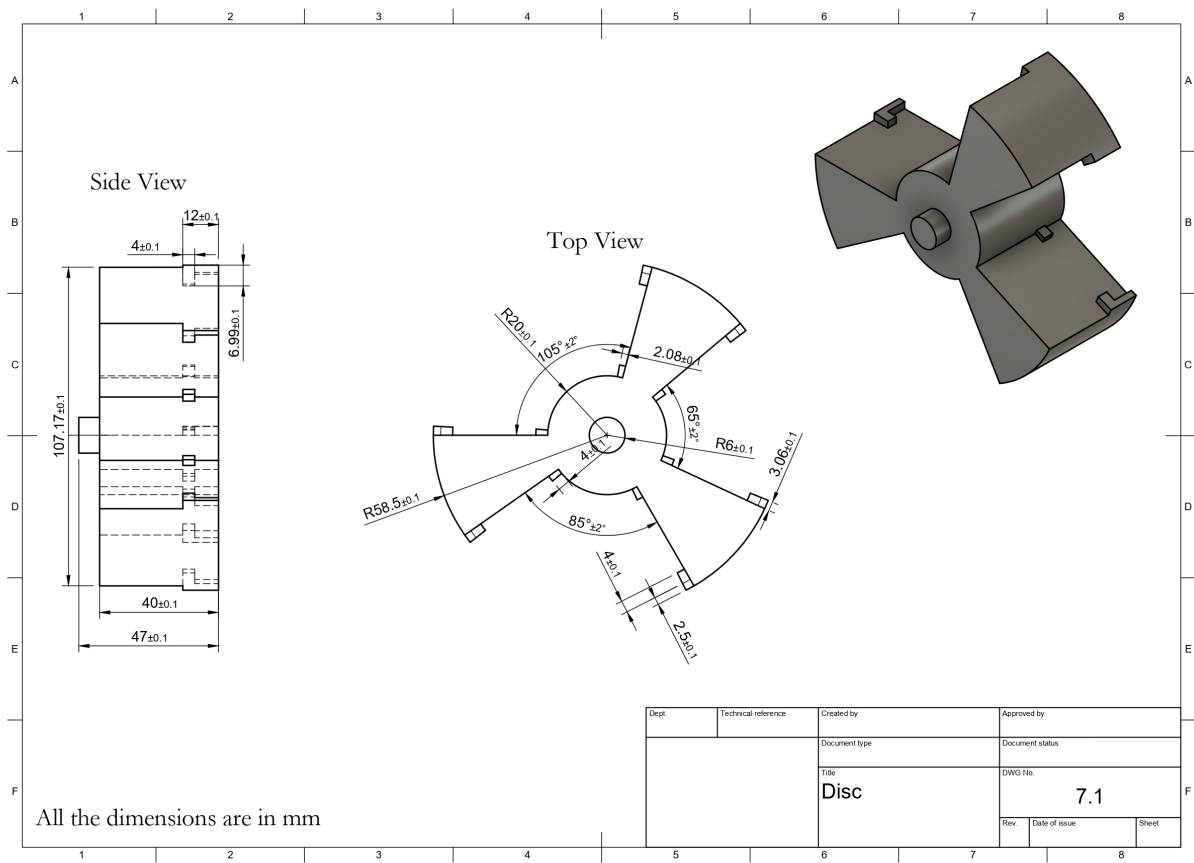


Figure 7: Disc(A)

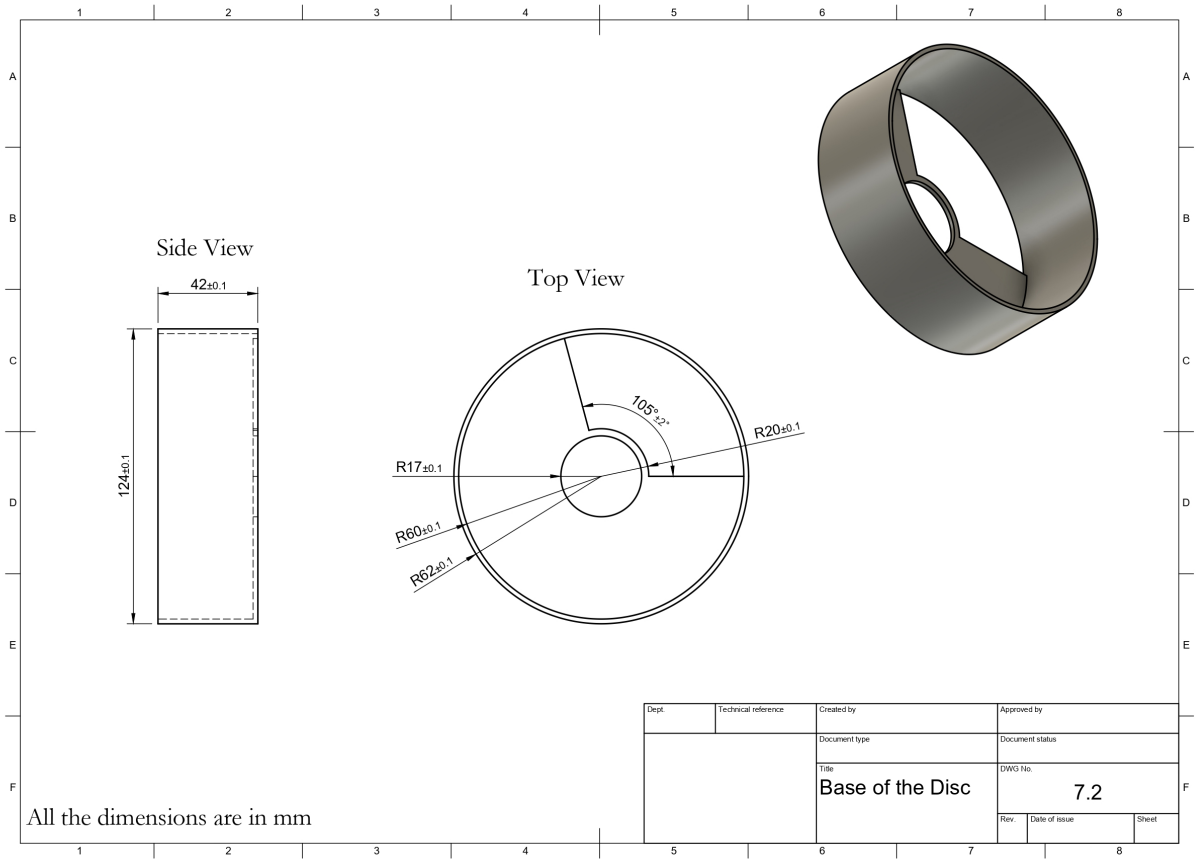


Figure 8: Disc(B)

3.3.3 Cost Analysis

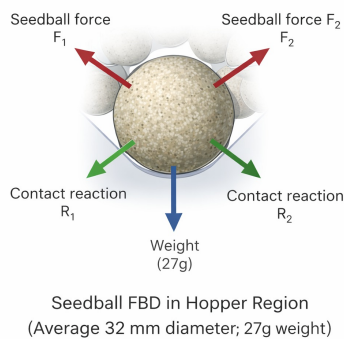
A detailed breakdown of the material and component expenses for the fabricated prototype is provided below.

Table 1: Cost Analysis

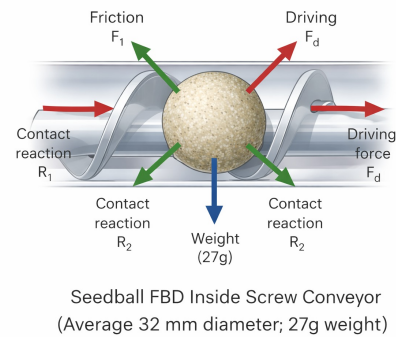
Item	Cost (Rs)
Carbon fiber filament	575
Acrylic Sheet	360
Microcontrollers	
Actuators (3)	
Motor Drivers	
Miscellaneous costs	
Total	

3.4 Free Body Diagrams

The free body diagrams of the seed-ball in hopper[Fig 9a] and horizontal auger[Fig 9b] are as follows :



(a) In Hopper



(b) In Auger

Figure 9: Free Body Diagrams

3.5 Control Logic

During operation, seedballs flow from the hopper aided by continuous mechanical agitation to prevent stagnation. They are then metered by a horizontal screw conveyor actively monitored by a TMC2209 stepper driver. If jamming occurs, integrated stall detection triggers an automated anti-jam routine, reversing the motor up to three times to clear the blockage before safely halting the system. Upon exiting the screw, seedballs are sequentially caught by a continuously rotating dispensing disk featuring conical radial grooves with varying incident angles. This geometry dynamically accommodates dimensional variations, ensuring a controlled and repeatable release through a single drop point

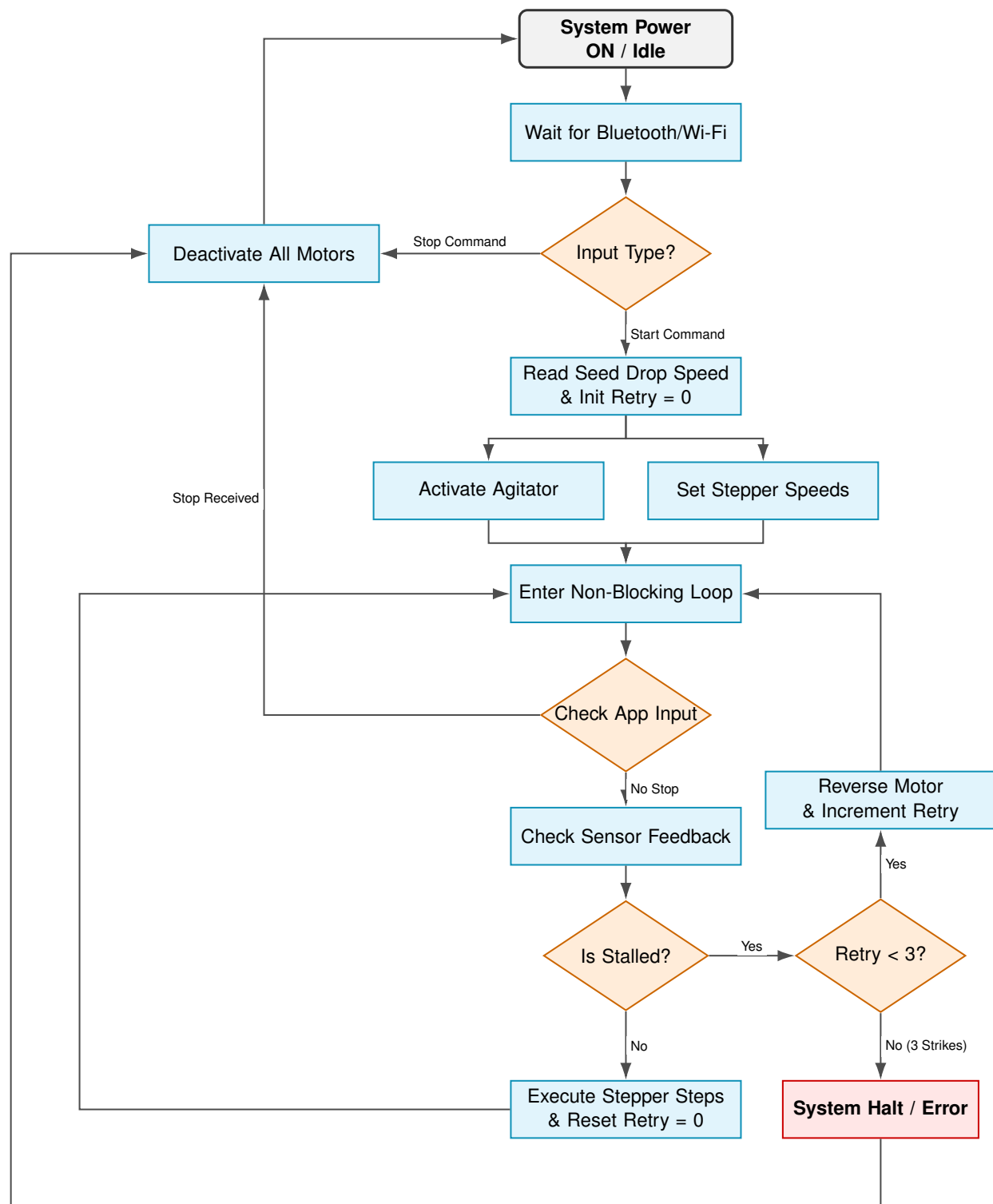


Figure 10: Control Flow Logic of the Seedball Mechanism.

3.6 Reliability: Blockage Points & Resolution Logic

3.6.1 Jamming and Failures

The potential blockage (jamming) points are as follows:

- (a) **Hopper throat:** Arching/bridging, ratholing, interlocking, overloading, moisture clumping, wall adhesion, dead zones.
- (b) **Agitator:** Peripheral dead zones, jamming.
- (c) **Hopper Outlet:** Wedging at lip/screw, misalignment, screw backpressure.
- (d) **Screw Conveyor:** Jams in flights, crushing/lodging, no advance, fill/speed issues, foreign particles, discharge buildup.
- (e) **Vertical Pipe:** Direction changes, bouncing, undersized diameter, stacking, electrostatic adhesion.
- (f) **Disc Assembly:** Seedball wedging in angular grooves, screw-to-disk timing mismatches, debris accumulation, and housing friction.
- (g) **Drone/Operational:** Acceleration/deceleration, tilted flights, vibrations, motor starts/stops, resonance, expansion.
- (h) **General:** Defective seedballs, dust, compaction, partial jams.

3.6.2 Detection

The system uses the TMC2209 driver's built-in stall detection feature. It monitors the load variations for motor and can sense when resistance suddenly increases an indication that a jam may be forming allowing the system to respond quickly and prevent further blockage.

3.6.3 Solutions

To mitigate blockages, the system employs a combined hardware-software strategy. Mechanically, continuous hopper agitation prevents initial material bridging, while the dispensing disk's varied radial grooves safely accommodate different seedball sizes without lodging. If a jam occurs in the screw conveyor, the TMC2209 driver detects the resistance spike via its integrated stall sensing. The control logic immediately triggers an automated anti-jam routine, reversing the motor to clear the obstruction. The system will attempt this up to three times; if successful, normal operation resumes and the retry counter resets. If the jam persists after three consecutive attempts, the system safely halts to prevent hardware damage.

3.7 Future Scope

Future developments will focus on enhancing dispensing precision by further minimizing the likelihood of multi-seedball release through improved metering strategies and more precise control mechanisms. Additionally, the integration of GPS-based dispersal logic is envisioned to enable location-specific, mission-aware deployment of seedballs. Expanding beyond single-unit operations, future control software could enable multi-drone swarm synchronization for rapid, large-scale ecological restoration.

An issue which has not yet been solved is the automation of maintenance. The **accumulation of earth** is inevitable and handling it will require some kind of rinsing mechanism which is difficult.

3.8 Mechanism Summary

This section summarizes the overall performance, strengths, and limitations of the mechanism, providing a concise evaluation of its feasibility and practical implementation.

3.8.1 Overall feasibility statement

The proposed seedball dispersal mechanism is feasible within the mass, dimensional, and power constraints of a drone mounted payload. Its hybrid use of gravity assisted flow and mechanically metered transport enables

reliable handling of non-uniform seedballs. With appropriate testing and calibration, the system is well-suited for autonomous aerial deployment.

3.8.2 Key Strengths

The major advantages of the mechanism are its ability to tolerate variations in seedball size, its multi stage jam prevention by means of agitation along with stall detection for motor reversal, and a modular design which makes it easier to maintain and upgrade in the future. In addition, sensor based monitoring helps to enhance the overall reliability of the machine during autonomous operation.

3.8.3 Risks and Limitations

The main limitations are that it is impossible to completely get rid of the risk of multi seedball release and the system is sensitive to mechanical tolerances in the screw and disc assemblies. Environmental factors like moisture and debris can also impact flow behaviour, which therefore requires controlled testing and refinement of the system.

4 Prototype Validation

4.1 Prototype Specifications and Scaling

To validate the mechanical logic and flow consistency of the proposed seedball dispersal payload, a physical prototype was fabricated at a 1:2 scale relative to the full-size design.

1. **Dimensional Scaling:** The original payload was designed to operate within maximum spatial constraints of 350 mm × 300 mm × 350 mm to fit standard drone-mounted parameters. The validation prototype successfully condensed this architecture into a compact 14 cm × 17 cm × 22 cm footprint, effectively halving the volumetric requirements while maintaining the spatial relationships between the hopper, horizontal auger, and dual-disc dispensing stages.
2. **Seedball Parameters:** To accurately test the gravity-fed dynamics and prevent false-positive jam testing, the prototype utilizes representative seedballs scaled down by approximately 50%. While the full-scale payload is designed to handle seedballs averaging 32 mm in diameter (varying from 15 mm to 35 mm), the validation testing relies on approximately 13 mm diameter test spheres. This ensures the geometric interaction between the seedballs, the rotational agitator, and the screw conveyor flights remains physically accurate to the original design philosophy.
3. **Material and Component Selection:** Transitioning from the theoretical design to a physical build required adapting the structural materials for rapid manufacturability and testing. The prototype's primary structural framework and housing utilize a combination of carbon fiber and acrylic sheets, balancing rigidity with weight reduction. The system integrates the core control electronics, including the microcontroller and motor actuation hardware, directly into this scaled-down envelope. This allows for real-world testing of the TMC2209 driver's stall detection capabilities and the system's automated fault response logic within a representative physical space.

4.2 Performance Analytics and Operational Metrics

5 Conclusion

The Horizontal Auger with Disc mechanism continues to be a real and mechanically strong solution for the controlled release of seedball drones in the air. This model solves the main issue of aerial reforestation, which is the regulated and even release of non uniform seedballs under the limiting conditions of weight, power, and reliability that come from deployment by a drone while at the same time minimizing the risk of jamming.

1. **Decoupled Material Flow**

The use of a horizontal screw conveyor as an intermediate buffer decouples the dispensing mechanism from the variable head pressure of the hopper. This isolation ensures predictable and metered material flow, reducing the likelihood of uncontrolled discharge and pressure-induced jamming.

2. **Adaptive Precision**

The Disc mechanism, incorporating conical openings of different angles made to accommodate different seed-ball dimensions, enables adaptive handling of seedballs with diameters ranging from 15 mm to 35 mm. This significantly reduces jamming risks commonly encountered in rigid, single-aperture systems and improves operational robustness.

3. **Simplified Diagnostic Feedback**

Jamming detection is achieved through monitoring motor stalling within the screw conveyor and disc assemblies. TMC2209 driver has stall detection to reliably detect stalling. Upon detection, the control logic can autonomously initiate temporary motor reversal without requiring external sensors.

4. **Mass and Energy Efficiency**

Combined with a lightweight carbon-fiber structural frame and direct power supply from the drone's onboard battery, the module remains within the 20 kg payload constraint.

5. **Future Scope**

While the current configuration provides a robust foundation for reliable seedball dispersal, future iterations will focus on enhancing singulation precision through active gating mechanisms, fail safe operations integrating and mission-aware GPS-based deployment logic.

6 References

- [1] K. To, P.-Y. Lai, and H. K. Pak, *Jamming of Granular Flow in a Two-Dimensional Hopper*, Phys. Rev. Lett. **86**, 71–74 (2001).